# Analyzing Graduation Admission Data Using Bayesian Beta Regression: A Probabilistic Approach


## GROUP 7

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# Introduction

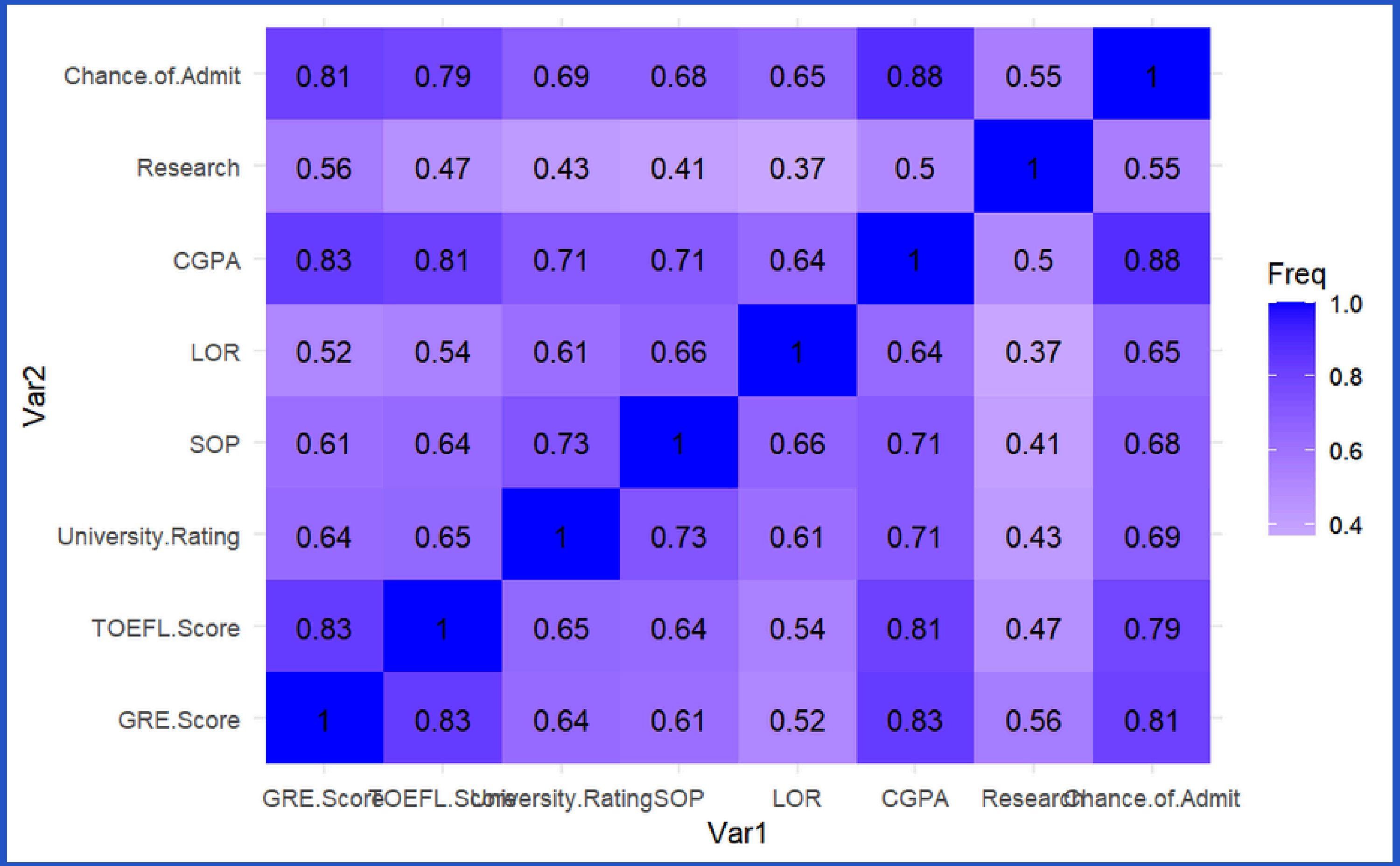
Education is a broad term encompassing knowledge, experience, learning, and teaching. As John Dewey stated, "Education is the process of living through a continuous reconstruction of experiences."

## DATASET OVERVIEW

**7 column with 500 student records**

**8 variables**

GRE.Score (integer), TOEFL.Score (integer), University.Rating (integer), SOP (numeric), LOR (numeric), CGPA (numeric), Research (binary), Chance.of.Admit (numeric)




**Target variable: Chance of Admit (bounded between 0 and 1)**

# Bayesian Beta Regression

well-suited for modeling continuous variables that are constrained to a specific interval, such as the **"Chance of Admit"**, it uses two parameters that are derived from the data's mean and precision, which are analogous to the mean and variance used in normal distributions.

The model was implemented in JAGS,

- normal priors **(Normal(0,10^-6))** for regression coefficients
- gamma prior **(Gamma(0.001,0.001))** for the precision parameter.

**Prior**

$$\boldsymbol{\beta}|\sigma^2 \sim Normal(\boldsymbol{\mu}, \sigma^2\boldsymbol{\Omega})$$

**Posterior**

$$Y_i|\boldsymbol{\beta}, r \sim Beta(rq_i, r(1-q_i))$$

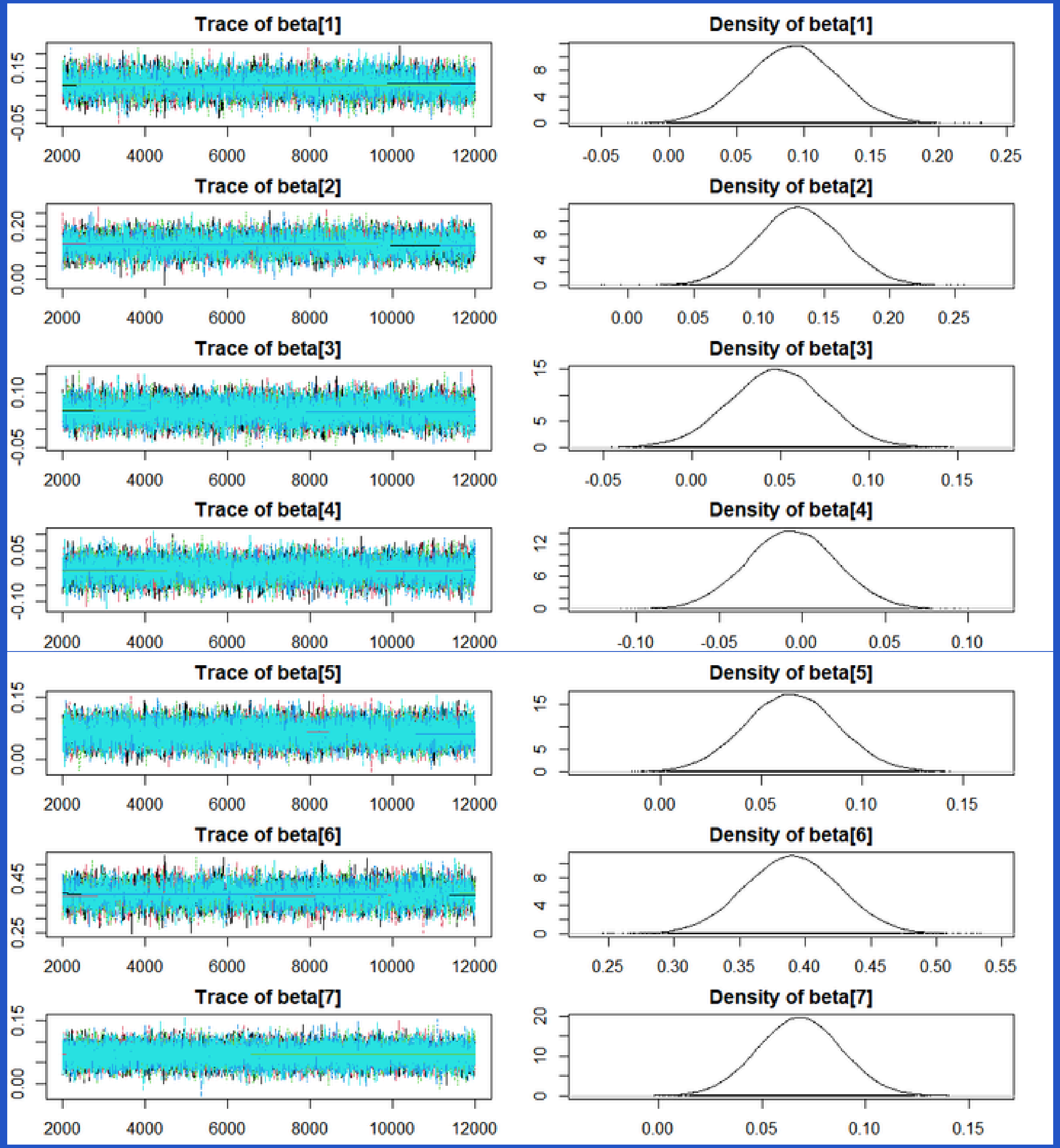
# JAGS MODEL

```
model_str <- "
  model{
    for(i in 1:N){
      Y[i] ~ dbeta(phi*mu[i], phi * (1-mu[i]))
      logit(mu[i]) <- beta_0 + inprod(beta[1:col], x[i, 1:col])
      log_lik[i] <- logdensity.beta(Y[i], mu[i] * phi, (1   mu[i]) * phi)
    }

    #Priors
    beta_0 ~ dnorm(0, 1e-6)

    for(i in 1:col){
      beta[i] ~ dnorm(0, 1e-6)
    }
    phi ~ dgamma(0.001, 0.001)
  }
"
```

GROUP 7

# RESULTS

```
Iterations = 2001:7000
Thinning interval = 1
Number of chains = 5
Sample size per chain = 5000

1. Empirical mean and standard deviation for each variable,
   plus standard error of the mean:

            Mean      SD  Naive SE Time-series SE
beta[1]  0.13120 0.03313 0.0002095      0.0006461
beta[2]  0.11360 0.03232 0.0002044      0.0006016
beta[3]  0.04489 0.02530 0.0001600      0.0003467
beta[4] -0.01406 0.02583 0.0001634      0.0003828
beta[5]  0.08179 0.02249 0.0001423      0.0002871
beta[6]  0.37348 0.03354 0.0002121      0.0006425
beta[7]  0.04218 0.01956 0.0001237      0.0002122
beta_0   1.06103 0.01667 0.0001054      0.0001423
phi     50.80270 3.60696 0.0228124      0.0298421

2. Quantiles for each variable:

             2.5%      25%      50%      75%    97.5%
beta[1]  0.065163  0.10908  0.13123 0.153650  0.19596
beta[2]  0.050445  0.09215  0.11355 0.135171  0.17699
beta[3] -0.003968  0.02769  0.04485 0.061881  0.09463
beta[4] -0.064201 -0.03138 -0.01405 0.003252  0.03622
beta[5]  0.038101  0.06649  0.08174 0.097036  0.12593
beta[6]  0.308166  0.35090  0.37305 0.395773  0.44039
beta[7]  0.004180  0.02909  0.04198 0.055347  0.08072
beta_0   1.028574  1.04978  1.06105 1.072149  1.09375
phi     44.011474 48.26381 50.71266 53.211603 57.99493
```



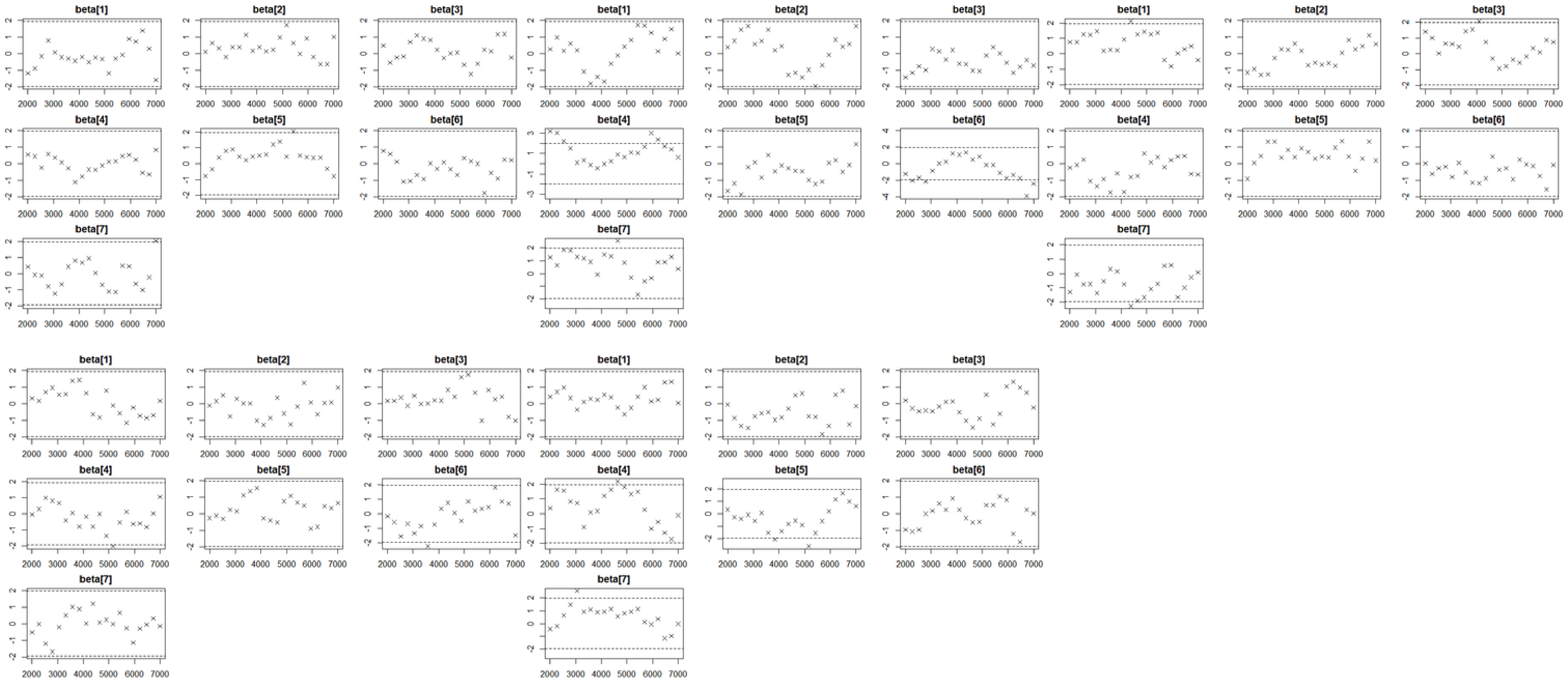

**CGPA and GRE** scores have the most positive effects on admission chances, with mean values of 0.37348 and **0.1312** and credible intervals that exclude zero

**successfully converged,** model has stabilized and is providing reliable estimates, particularly for the key predictors of the chance of admission

# CONVERGENCE DIAGNOSTIC

# Geweke Diagnostic

```
Chain 1
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7]  beta_0     phi
-1.7378  1.6426 -2.4203 -0.4089  0.4455  0.8929  2.0861 -0.9663 -0.3063

Chain 2
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7]  beta_0     phi
 0.5615 -1.0057  0.9255 -1.4266  0.3220  0.5747 -0.5469 -1.3705  0.9675

Chain 3
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

 beta[1]  beta[2]  beta[3]  beta[4]  beta[5]  beta[6]  beta[7]   beta_0      phi
-0.70905 -0.78897 -0.53825 -0.05437 -1.01871  1.89991  0.56635 -0.84314 -1.25545

Chain 4
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7]  beta_0     phi
 0.7101 -0.5322  0.2100  0.6352 -0.7630 -0.4713  0.4204  0.4869 -0.2498

Chain 5
Fraction in 1st window = 0.1
Fraction in 2nd window = 0.5

beta[1] beta[2] beta[3] beta[4] beta[5] beta[6] beta[7]  beta_0     phi
 0.4121 -0.5882  1.3565 -0.3680  0.3430 -0.2404 -0.5204  0.9498  1.0835
```



assume that in the first chain **beta3 and beta7** is over our threshold and is assumed to be divergent.

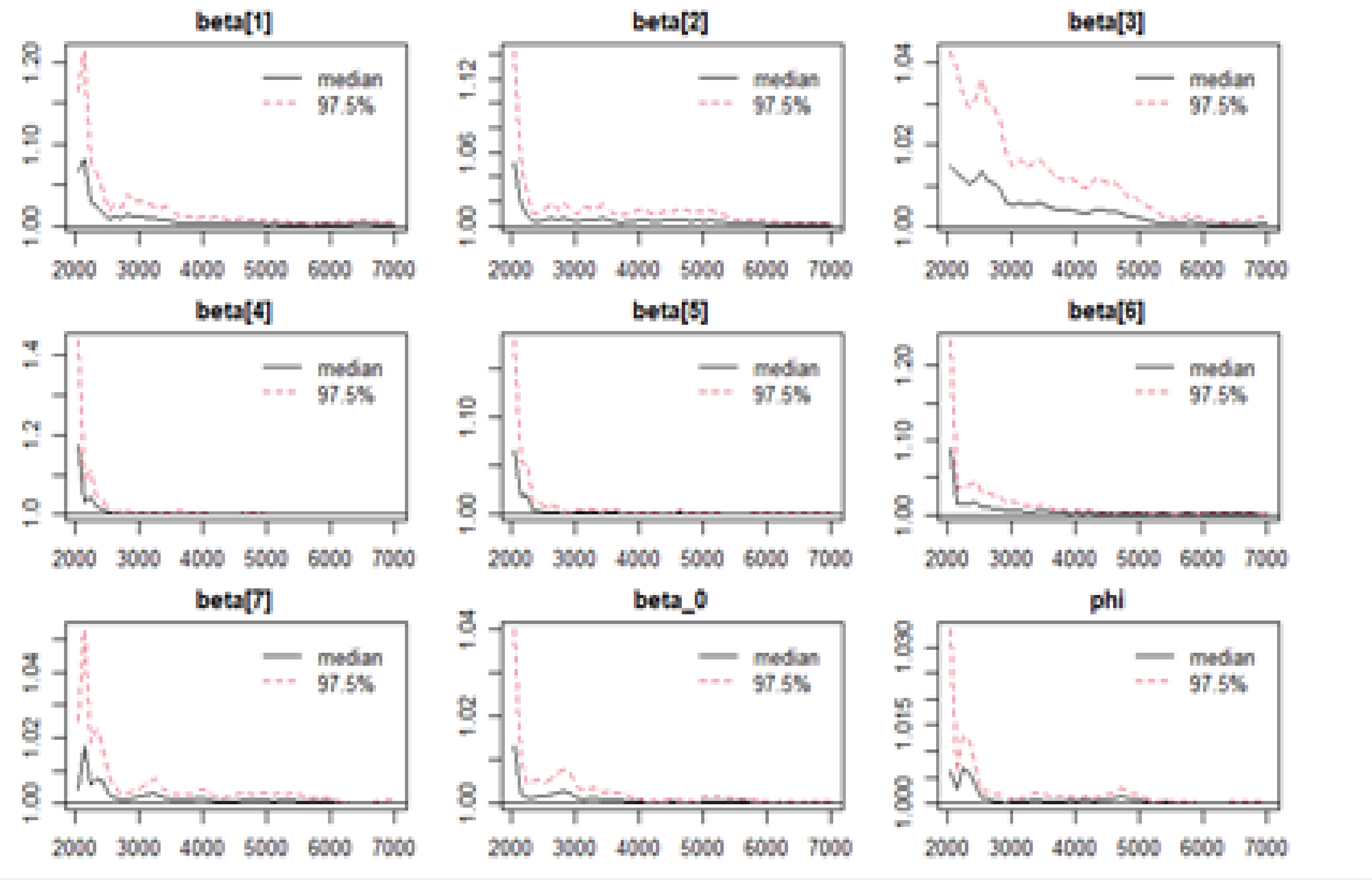
Chain 1 exceed the aforementioned boundaries, specifically for (**University.Rating for β3 and Research for β7**). This indicates that the values for the corresponding slopes **did not converge**

Provides a standardized Z-score, where values far from zero may indicate that the chain has not yet converged to its stationary distribution.

$$Z = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{Var(x_1) + Var(x_2)}}$$

# Gelman-Rubin Diagnostic

```
Potential scale reduction factors:

        Point est. Upper C.I.
beta[1]          1          1
beta[2]          1          1
beta[3]          1          1
beta[4]          1          1
beta[5]          1          1
beta[6]          1          1
beta[7]          1          1
beta_0           1          1
phi              1          1

Multivariate psrf

1
```




This result strongly indicates that the chains have achieved **perfect convergence**

$$\hat{R} = \frac{\hat{V}}{W}$$

R = 1 indicates perfect convergence as the variability within each chain (W) is similar to the variability across chains (V).

# Raftery-Lewis Diagnostic

| | Burn-in (M) | Total (N) | Lower bound (Nmin) | Dependence factor (I) |
|---|---|---|---|---|
| beta[1] | 15 | 18558 | 3746 | 4.95 |
| beta[2] | 14 | 15042 | 3746 | 4.02 |
| beta[3] | 7 | 7819 | 3746 | 2.09 |
| beta[4] | 7 | 8197 | 3746 | 2.19 |
| beta[5] | 7 | 7675 | 3746 | 2.05 |
| beta[6] | 14 | 14182 | 3746 | 3.79 |
| beta[7] | 6 | 6406 | 3746 | 1.71 |

| | Burn-in (M) | Total (N) | Lower bound (Nmin) | Dependence factor (I) |
|---|---|---|---|---|
| beta[1] | 15 | 17133 | 3746 | 4.57 |
| beta[2] | 12 | 14950 | 3746 | 3.99 |
| beta[3] | 12 | 14096 | 3746 | 3.76 |
| beta[4] | 9 | 9307 | 3746 | 2.48 |
| beta[5] | 7 | 7819 | 3746 | 2.09 |
| beta[6] | 14 | 15054 | 3746 | 4.02 |
| beta[7] | 7 | 7262 | 3746 | 1.94 |

| | Burn-in (M) | Total (N) | Lower bound (Nmin) | Dependence factor (I) |
|---|---|---|---|---|
| beta[1] | 14 | 14470 | 3746 | 3.86 |
| beta[2] | 16 | 16564 | 3746 | 4.42 |
| beta[3] | 7 | 8119 | 3746 | 2.17 |
| beta[4] | 7 | 7893 | 3746 | 2.11 |
| beta[5] | 6 | 6939 | 3746 | 1.85 |
| beta[6] | 14 | 16312 | 3746 | 4.35 |
| beta[7] | 6 | 6294 | 3746 | 1.68 |

| | Burn-in (M) | Total (N) | Lower bound (Nmin) | Dependence factor (I) |
|---|---|---|---|---|
| beta[1] | 12 | 12556 | 3746 | 3.35 |
| beta[2] | 14 | 15296 | 3746 | 4.08 |
| beta[3] | 12 | 12786 | 3746 | 3.41 |
| beta[4] | 9 | 9691 | 3746 | 2.59 |
| beta[5] | 7 | 7465 | 3746 | 1.99 |
| beta[6] | 16 | 17850 | 3746 | 4.77 |
| beta[7] | 6 | 6520 | 3746 | 1.74 |

| | Burn-in (M) | Total (N) | Lower bound (Nmin) | Dependence factor (I) |
|---|---|---|---|---|
| beta[1] | 12 | 13244 | 3746 | 3.54 |
| beta[2] | 12 | 14306 | 3746 | 3.82 |
| beta[3] | 7 | 8197 | 3746 | 2.19 |
| beta[4] | 7 | 7819 | 3746 | 2.09 |
| beta[5] | 7 | 7893 | 3746 | 2.11 |
| beta[6] | 12 | 14908 | 3746 | 3.98 |
| beta[7] | 6 | 6695 | 3746 | 1.79 |

Dependence Factors (I) for the majority of parameters fall within the **range of 2 to 4**

The degree of autocorrelation present in the Markov Chain Monte Carlo (MCMC) chain, which affects the efficiency of the sampling process (represented by I).

- I 1 suggests little to no autocorrelation
- I between 2 and 4 suggests moderate autocorrelation
- I > 5 suggests significant autocorrelation

# Effective Sample Size

```
  beta[1]   beta[2]   beta[3]   beta[4]   beta[5]   beta[6]   beta[7]    beta_0
 2652.491  2898.796  5370.582  4557.030  6138.848  2751.610  8513.012 13772.620
      phi
14622.371
```

A higher ESS indicates better sampling efficiency and more reliable parameter estimates.

Some parameters, like **βo and ϕ** have a comparatively large effective sample size. These variables specifically have **little to no autocorrelation.**

# Autocorrelation

**Chain 3** achieving decorrelation at a slightly higher lag compared to the others

```
Autocorr is 0 for chain 1 at lag -18
Autocorr is 0 for chain 2 at lag 22
Autocorr is 0 for chain 3 at lag -29
Autocorr is 0 for chain 4 at lag -13
Autocorr is 0 for chain 5 at lag -18
```

# EVALUATION MODEL

# WAIC (Watanabe-Akaike Information Criterion)

Comparison between the Beta regression model and itself highlights WAIC (Watanabe-Akaike Information Criterion) shows a score of **-1118.597**, with an effective complexity penalty (p-WAIC) of **9.435902**.

**A value of 9.435902 suggests the model has moderate complexity, balancing flexibility and generalizability.**

# LOO-CV
# (Leave-One-Out Cross-Validation)

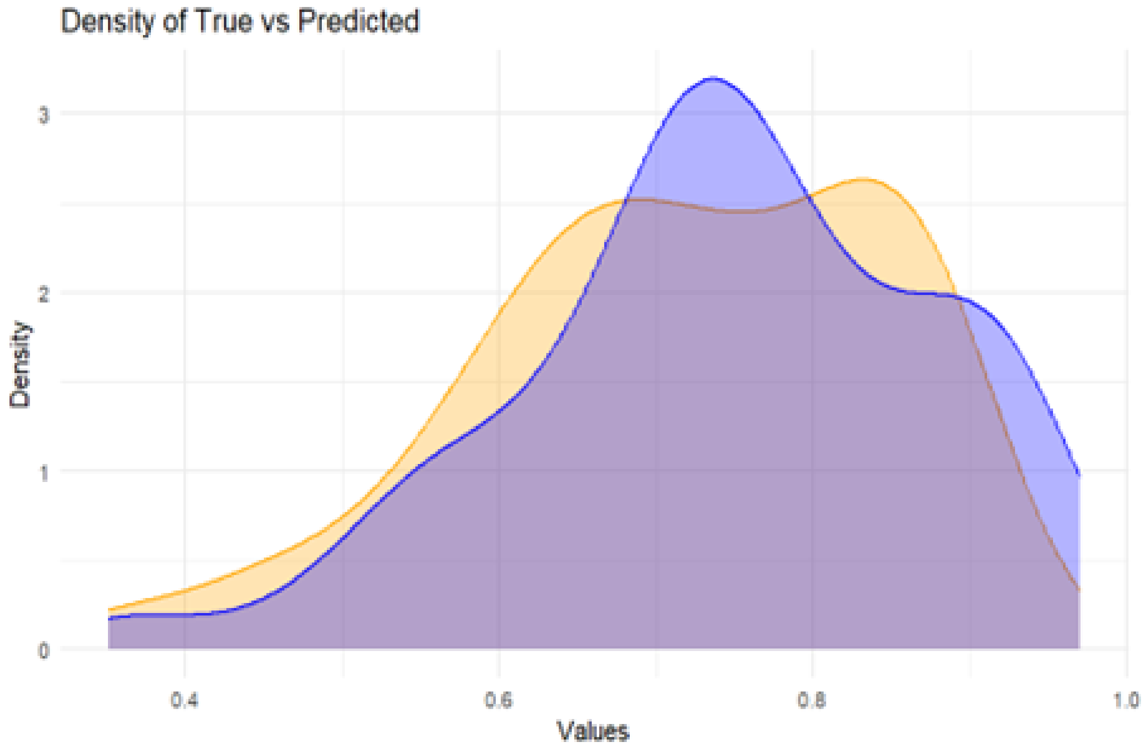
| | Estimate <S3: AsIs> | SE <S3: AsIs> |
|---|---|---|
| elpd_loo | 708.8 | 20.9 |
| p_loo | 11.1 | 1.1 |
| looic | -1417.6 | 41.9 |

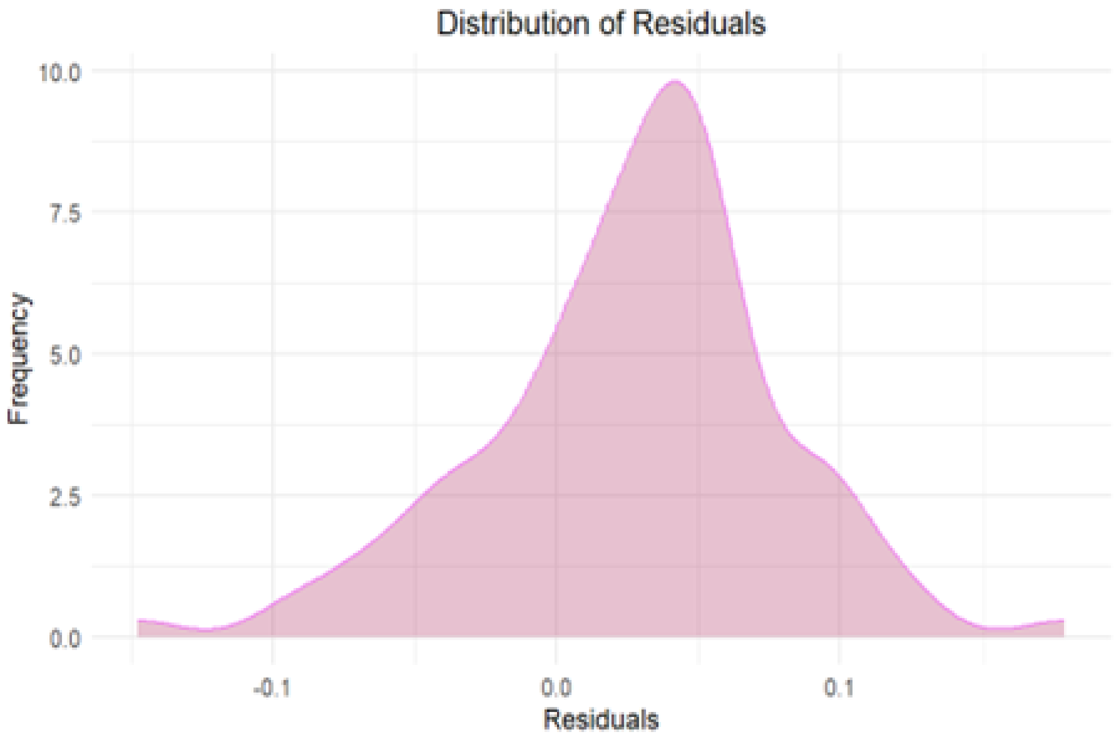
- The **Expected Log Predictive Density** (elpd_loo) is **708.8** with a **Standard Error** (SE) of **20.9**, reflecting the model's predictive accuracy.
- The **Effective Number of Parameters** (p_loo), which measures model complexity, is **11.1 with an SE of 1.1**, indicating the effective degrees of freedom used by the model while accounting for potential overfitting. **LOO**
- **Information Criterion** (LOOIC), a value of **-1417.6** with an **SE of 41.9**, where lower values signify better model performance.

# POSTERIOR PREDICTIVE CHECKS

# Graphical Representation

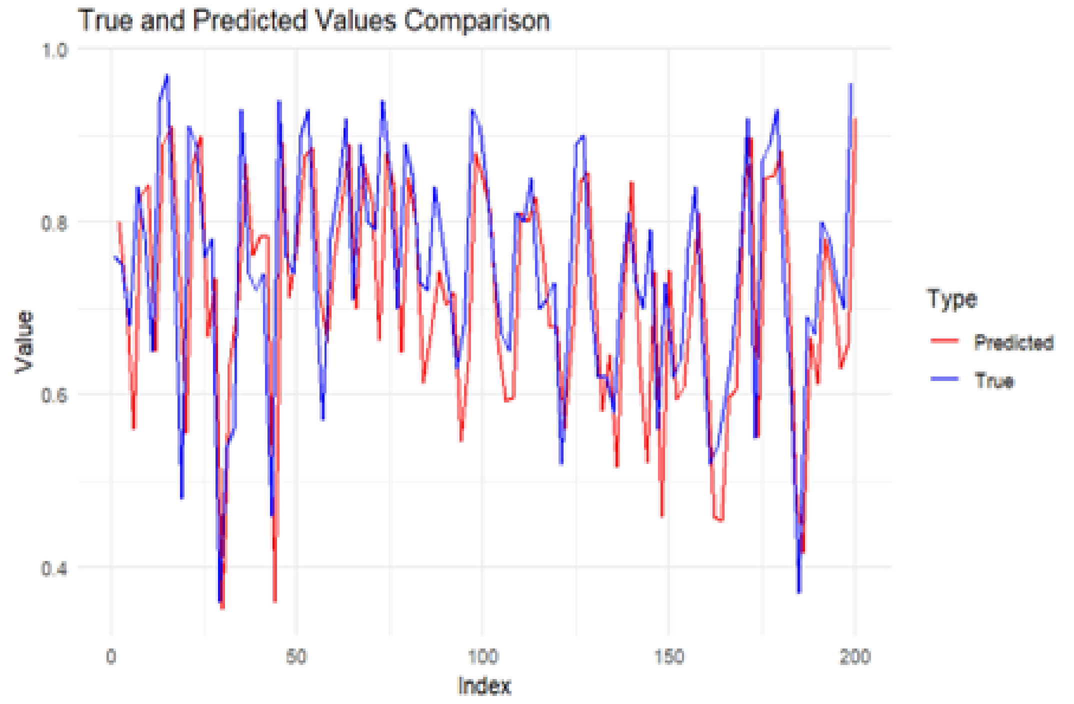
| True <dbl> | Predicted <dbl> | Residual <dbl> |
|---|---|---|
| 0.76 | 0.8003547 | -0.0403546583 |
| 0.75 | 0.7129708 | 0.0370292117 |
| 0.68 | 0.5599991 | 0.1200008793 |
| 0.84 | 0.8302064 | 0.0097935941 |
| 0.78 | 0.8428679 | -0.0628679250 |
| 0.65 | 0.6506469 | -0.0006469102 |
| 0.94 | 0.8900826 | 0.0499174298 |
| 0.97 | 0.9107017 | 0.0592983444 |
| 0.76 | 0.7674493 | -0.0074493221 |
| 0.48 | 0.5546542 | -0.0746542271 |




$$p(\tilde{y} \mid y) = \int p(\tilde{y} \mid \Theta) p(\Theta \mid y) d\Theta$$

ϴ denotes the model parameters, p(ỹ | ϴ) represents the likelihood of the simulated data given the parameters, and p(ϴ | y) corresponds the posterior distribution of the parameters given the observed data y



# Mean Squared Error (MSE)

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \tilde{y}_i)^2$$

Measure of the average squared differences between the observed values (y) and the predicted values (ỹ). A **lower MSE** indicates a better fit, as it demonstrates that the **predicted values are closer to the observed data**

| Model | Baseline |
|---|---|
| [1] 0.003412774 | [1] 0.0179975 |

# Posterior-Predictive p-value (ppp-value)

**0.45**

$$p_{post} = P(T(\tilde{y}) \geq T(y) \mid y)$$

Values close to **0.5** indicate a **good match** between the simulated and observed data, while **extreme values (near 0 and 1)** signal potential model **misspecifications**

# Comparison with Existing Literature

| Existing Literature | Proposed Model |
|---|---|
| Applied Bayesian Model Averaging | Due to us only using one model we did not apply Bayesian Model Averaging |
| Only used Mean Squared Error as the evaluation metric | Used Posterior-Predictive p-value (ppp-value) as well as Mean Squared Error (MSE) as evaluation metrics |
| A simpler model resulting in a higher MSE score showing the model's is less accurate compared to the proposed model | A more complex yet more resource-intensive model resulting in a lower MSE score (0.003406591) demonstrating a more accurately predicting model. |

# Conclusion

This study aims to properly evaluate a student's chance of admission to a selected university by using Bayesian methods. Among these methods, Bayesian Beta Regression is utilized to model the probability of admission, as it is particularly suited for analyzing outcomes that fall within a bounded range, such as probabilities or proportions.

The analysis successfully demonstrated the applicability of **Bayesian Beta Regression** in modeling admission probabilities for university applicants. The Bayesian Beta Regression model demonstrated robust predictive accuracy, as evidenced by the **Mean Squared Error (MSE) of 0.003406591** and flexibility in handling bounded outcomes, as shown by the Posterior Predictive p-value **(PPP-value) of 0.45.**

# THANK YOU

## GROUP 7

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